Break-even and the price decision

Teacher guidance and answers | Original TBS classroom practice

Use this lesson after students have met fixed and variable costs, contribution and the break-even formula. Students move from calculating to judging: two plans, one set of costs, and a decision that depends on assumptions. The tasks are planned to take about 55 minutes. Allow about 65 minutes in the timetable for handing out, whole-class feedback and the plenary. It has not been trialled in a classroom. Suitable for classes studying Edexcel 9BS0 Theme 2, in Year 12 or as Year 13 revision.

Curriculum fit

Pearson Edexcel A level Business 9BS0, Specification Issue 1 (September 2014), Theme 2, 2.2.3 Break-even: (a) contribution, (b) break-even point, (c) using contribution to calculate break-even, (d) margin of safety, (e) interpretation of break-even charts and (f) limitations of break-even analysis. 2.2.2(b) fixed and variable costs is the prerequisite. Checked against the archived Issue 1 specification; the live Pearson course page was opened on 8 October 2026 and showed no withdrawal notice in the fetched text, so please confirm the specification for your cohort with your exams officer. This is a TBS lesson, not a Pearson resource; it is not endorsed by Pearson and uses no Pearson exam questions or mark scheme.

All business details and numbers are fictional. The answers are indicative teaching guidance only. There is no official mark tariff, level descriptor or AO allocation for these tasks, and no grade claim is made.

Lesson plan (about 55 minutes of tasks, estimated)

Stage	Minutes	What happens
Starter: sort the costs	5	Students classify six costs, then compare.
Task 1: the business today	10	Contribution, break-even, profit, margin of safety. Pair check.
Task 2: read the chart	8	Read break-even and profit from the chart; confirm one value by calculation.
Task 3: compare two ideas	17	Complete the table, then test the forecasts.
Task 4: decide	10	Written recommendation with figures and one limitation.
Exit ticket	5	Two quick questions.
Total	55	

45-minute version: starter 5, Task 1 8, Task 2 6, Task 3 14, Task 4 7, exit ticket 5. Give the Idea A column to half the class and the Idea B column to the other half, then swap results before Task 4. Task 4 is tight at 7 minutes: ask for about 60 words, or set it as homework. This timing has not been trialled. The 45 minutes are task time only, so in a real 45-minute period, with handing out and feedback, move the exit ticket to homework.

Support and extension

Support: put contribution = price - variable cost and break-even = fixed costs / contribution on the board, and complete Task 1a with the class. Extension: at the current 800 candles and the current costs, what price per candle would make Ashby Candles exactly break even? The answer is in the extension note on page 4. Ask whether that is a sensible price.

Answers: starter and Task 1

Indicative guidance for teachers. This is not an exam board mark scheme and carries no marks or grades.

Workshop rent F | wax V | glass jars V | insurance F | website hosting F | postage paid per order V.

Postage per order is variable if the firm pays it for each order. Accept a reasoned alternative.

Task 1	Answer	Working
a. Contribution per candle	£7	12 - 5
b. Break-even	600 candles	4,200 / 7
c. Profit at 800	£1,400	800 x 7 = 5,600; 5,600 - 4,200
d. Margin of safety	200 candles; 25.0%	800 - 600; 200 / 800

Answers: Task 2 (chart)

a. Break-even is 600 candles at about £7,200 revenue (accept 550 to 650 candles and £6,600 to £7,800, since the chart is read by eye). The lines cross there; revenue and total costs are both £7,200.

b. At 400 candles there is a loss of about £1,400 (revenue £4,800, total costs £6,200). At 1,000 candles there is a profit of about £2,800 (revenue £12,000, total costs £9,200). Accept readings within about £500 of these values.

c. 1,000 x 7 = 7,000; 7,000 - 4,200 = £2,800. Revenue 12,000 - variable costs 5,000 - fixed costs 4,200 gives the same result.

Answers: Task 3

Indicative guidance, not an exam board mark scheme. The Today column of the student table is already filled in with the Task 1 answers, so students can check their Task 1 work before comparing the ideas.

£ unless stated	Today	Idea A	Idea B
Contribution per candle	7	5	7
Fixed costs per month	4,200	4,200	4,800
Break-even output (candles)	600	840	686 (accept 685.7)
Expected sales (candles)	800	1,200	950
Expected monthly profit	1,400	1,800	1,850
Margin of safety (candles)	200	360	264
Margin of safety (% of expected sales)	25.0%	30.0%	27.8%

Working and tests

Idea A: contribution is 10 - 5 = £5. Break-even is 4,200 / 5 = 840 candles. Profit at 1,200 candles is 1,200 x 5 - 4,200 = £1,800. Margin of safety is 1,200 - 840 = 360 candles, which is 30.0% of 1,200.

Idea B: fixed costs are 4,200 + 600 = £4,800. Contribution stays £7. Break-even is 4,800 / 7 = 685.7, rounded up to 686 candles. Profit at 950 candles is 950 x 7 - 4,800 = £1,850. Margin of safety is 950 - 686 = 264 candles, which is 27.8% of 950 (this lesson rounds break-even up to whole candles, so the margin of safety is the whole number 264).

Test	Answer	Working
a. Idea A, sales of 900	£300 profit	900 x 5 = 4,500; 4,500 - 4,200
b. Idea A, sales needed for £1,400 profit	1,120 candles	(4,200 + 1,400) / 5; a 40% rise on 800
c. Idea B, no extra sales	£800 profit	800 x 7 = 5,600; 5,600 - 4,800

Useful follow-up: if Idea A sells only 800 candles, it makes a loss of £200 (800 x 5 - 4,200). On these figures Idea B carries less risk, because its break-even (686 candles) is below today's sales and test c still gives a £800 profit. It would make a loss below 686 candles, so it is not risk free.

Task 4: indicative decision guidance

A well supported answer picks one option, quotes figures and compares them. Likely strong lines: the two ideas are close on expected profit (£1,850 for B and £1,800 for A) but differ in risk. Idea A has the larger margin of safety (360 candles, 30.0%, against 264 candles, 27.8%), a bigger cushion once the extra sales arrive, but it needs sales to rise by 400 candles (50%) and would make a loss of £200 if sales stayed at 800; it needs a 40% rise (to 1,120 candles) just to match today's profit. Idea B needs a rise of only 150 candles (18.8%) and still earns £800 if sales do not move. A student could back B for lower risk, A because a lower price could win new customers and builds a bigger cushion if demand responds, or neither until demand evidence is collected. All are acceptable if the figures are used correctly.

Limitations students may use, matching 2.2.3(f): the model assumes every candle is sold; it assumes the price and variable cost per candle do not change; fixed costs may not stay fixed at higher output; both sales forecasts are only estimates; competitors may react to a price cut; it ignores the effect on other products or on brand image. Look for a limitation tied to the case, not a pasted list.

Look for correct figures, a clear comparison, a limitation tied to the case and a conclusion that depends on the assumptions. Do not award a grade or percentage from these tasks.

Common errors: forgetting to round break-even up; dividing the margin of safety by break-even instead of by expected sales; using revenue instead of contribution; treating the extra advertising as a variable cost; assuming a break-even point proves a plan will make a profit.

Answers: exit ticket and extension

a. Break-even shows only the output at which revenue equals total costs. Profit depends on whether sales actually exceed it, and sales are forecast; the assumptions about price, costs and demand may not hold.

b. 3,500 / 7 = 500 candles.

Extension note: price = 4,200 / 800 + 5 = £10.25 per candle. Below that price, 800 candles would make a loss.

Sources and limits

Curriculum source: Pearson Edexcel Level 3 Advanced GCE in Business specification, Issue 1, September 2014, Theme 2, section 2.2 (printed page 15). The specification verifies the topic scope only. The tasks, figures, timings and indicative guidance are original TBS teaching design.